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10

Assessing Multimodal Listening

Mari Carmen Campoy-Cubillo

Mercedes Querol-Julián

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10.1 Introduction

This chapter studies the use of multimodal texts as input for listening comprehension assessment. It is oriented towards the analysis of criteria of multimodal listening tasks that should be met in the use of videotexts and other multimedia tools for assessing listening in foreign language teaching.

The majority of the studies investigating the use of video and audio-only resources in SL/FL listening focus on the comparison of students' performance when using one input or the other to understand a text (Brett, 1997; Gruba, 1993; Coniam, 2001; Wagner, 2008, 2010b). However, this type of comparison cannot determine whether one mode is better than another to enhance performance. In fact, students are responding to different means of information and different informational contents when using one or the other. Moreover, results could also vary according to the kinds of questions or output that are elicited from the tasks that are used. A correct answer to these questions may or may not be enhanced by one mode (audio recordings), or by the use of co-occurring modes (video recordings, i.e. audio and dynamic visual elements/nonverbal information), depending on whether the data provided by the mode(s) contains qualitative information for the requested answer. For instance, if we choose to include a video, does the nonverbal information aid comprehension? Are the questions clearly related to this nonverbal information? Besides, most studies on the use of video in the classroom also lack well-defined assessment criteria, which are an essential aspect in the learning process.

Regarding text type and nonverbal item testing, Brett (1997), for instance, compared different formats (audio, video, multimedia) with the same or similar text. However, the task type in this study mainly focused on form (understanding phrases and sentences, or deleted words to be written while listening) and paid no attention to nonverbal information. The results of this study showed that students performed better when a multimedia environment was used. This

suggested that a media input was better for the tasks assigned. However, if no attention is paid to other modes different from speech, then questions related to gestures, contextual images, etc. are not considered and therefore not tested. Thus, the use of video does not make any sense and it can be considered distracting.

The point is that the use of aural-only texts or videotexts should be determined by the aims of the listening activity. Thus, if our main aim is for our students to listen in order to recognize words and insert them in a fill in the gap exercise, and we want them to exercise their ability to recognize specific phonemes (phonemic discrimination), then audio-only texts might prove better than videos. In this case, we are not as interested in text comprehension as in phoneme and word recognition, and the students will not have any visual distraction (though this may not always be the case: lip movement for instance may also help listeners as observed in McGurk and MacDonald, 1976). However, if we aim at fostering multimodal comprehension skills, a broader perspective needs to be considered. Our students should be trained to understand and interpret situations where both linguistic and nonlinguistic features (gestures, facial expression, intonation, word stress, pauses, hesitations, etc.) co-occur and interact to create meaningful messages. For example, students may be asked to figure out how facial expressions or gestures can influence the linguistic message or even convey additional meaning.

This chapter intends to put forward a theoretical framework for the description of the kind of information that is relevant in each mode (e.g. gestures, facial expression, images, etc.), and the relationship between their linguistic and nonlinguistic features from a pedagogical point of view. It is by offering this theoretical description while considering Rost's (2011) general map of listening ability that we may provide the necessary foundation for understanding how to assess multimodal listening tasks.

10.2. Research on listening: paving the way towards multimodality

Listening is an essential skill in many areas of our lives. As Shellenbarger (2014) explained, "[t]he failure to listen well not only prolongs meetings and discussions but also can hurt relationships and damage careers." In an academic context, listening also has a relevant function since understanding information transmitted in the classroom is of paramount

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importance. Two main listening types can be distinguished: transactional or one-way listening, and interactional or two-way listening (cf. Lynch, 2011). The same event may contain both types of listening situations, as in many university lectures where this academic genre is understood as "a relatively informal conversational style of speech with no overt interaction between lecturer and audience" (Sueyoshi and Hardison, 2005, p.68-69). Research on interactional listening plays an essential role in higher education, for instance in tutorials or seminars, and in conversational style lectures (interactive), where empirical evidence has supported the benefits of interaction for listening comprehension in FL students (Morell, 2004, 2007). Transactional listening (e.g., listening to noninteractive lectures) has paid special attention to the way learners cope with taking notes (Badger, Sutherland, White & Haggis, 2001) and understand discourse structure (Tauroza & Allison, 1994), or how they recall information (Jung, 2003). However, even in more noninteractive listening situations, when the students can access the speaker, there may be a degree of interaction. It is transactional listening that we pay most attention to in this chapter, as we discuss the use of multimodal tools, and specifically videotexts, for teaching and assessing listening where there is no verbal interaction with the speaker.

The target groups that can benefit from this approach are FL students. It could also be of interest to SL students, though we pay more emphasis on the needs of FL because the experience that they have with the new language and culture is much lower. This position is justified by the experimental study done by Sueyoshi and Hardison (2005), who claimed that the experience in the SL dramatically influences the students' awareness and inclusion of visual cues in listening strategies.

There is no question that oral communication is multimodal, that is, speech is just one component part of the great amount of oral and visual information that is conveyed and perceived when we construct meaning (Jewitt, 2013), e.g. sound, music, background noise, gestures, facial expression, body posture, gaze, text, images, figures, etc. With this in mind, when teaching a FL, and particularly when teaching listening skills, it seems obvious that multimodal communication, and thus multimodal resources, should play a significant role. If we train our students to be communicatively competent in the FL, we need to give them the opportunity to learn from real life situations. Though the issue of authenticity is quite controversial in language learning, we cannot deny that "authentic language is the target of

virtually all language learners" (Rost, 2011, p. 165). Authenticity in language learning refers to many aspects and has many faces, e.g. the situation, the task, the language or the input. Thus far, authentic input has been generally characterized by spontaneous spoken discourse. Our stance in the definition of genuine listening input takes a broader perspective to embrace not only oral features, but also visual ones. Nowadays, Internet facilitates access to resources for the FL classroom, authentic multimodal input among them. The acknowledgment of the important role of this type of input leads to a (re)definition of what is understood by listening.

The International Listening Association defines listening as "the process of receiving, constructing meaning from and responding to spoken and/or nonverbal messages" (ILA, 1995, p.4). As can be seen in the definition, the nonverbal part of the listening process is included. However, tradition in language teaching has always put more emphasis on grammar and lexis than on the interactional and nonverbal aspects of communication. Likewise, more importance is given to understanding specific words or specific information than to being able to understand the message as a whole. This is reflected in the fact that most listening tasks do not include questions on meaning inference, or on the understanding of interactional, nonverbal or contextual information.

From a historical point of view (Ariza & Hancock, 2003) the different approaches to teaching and assessing listening have evolved in relation to how the listening process was understood within the language learning process. In the environmentalist approach, the role of the listener was to simply listen for structures; learners were required to repeat, imitate and memorize isolated words and phrases. This role was broadened in the innatist approach where the listener should participate in the listening process to understand what is being said. The importance here lies in the reception of the message. This participation became a more important part of the construct from the interactionalist approach point of view. In this case, participation is more active since students were required to listen with a purpose, thus the listener's role is to listen both for content and meaning. The listener is asked to process the information and to construct meaning, and to activate content (cultural knowledge, topic familiarity, previous experience, etc.) and formal schemata (rhetorical conventions, discourse forms, and text types). Other components of the listening construct became more visible in the interpretive approach, such as pragmatic or sociolinguistic issues, which focused not only on processing information, but also on being able to interpret it within a communicative

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context. Thus, information is constructed alongside the listening situation and is modified as more information is revealed and (re)interpreted. Likewise, a quest for input authenticity was encouraged. Top-down and bottom-up approaches to listening (Buck, 2001) focus on the learners' prior knowledge to make predictions and on the recognition of linguistic elements respectively. In many cases both are used in a language classroom as they also are in real life.

However, to include all the aspects that have been incorporated as part of a listening event along the years, the interpretation of both verbal and nonverbal information is also needed. This is where the multimodal approach we propose comes in. This approach requires being aware of the modes that are activated in a communicative event to generate meaning. It requires the learner to develop an ability to observe how verbal and nonverbal cues co-construct meaning. It means teaching and learning how these modes operate, and designing models of possible mode interaction or co-occurrence. When designing materials for the classroom and assessment tasks, the multimodal approach needs to consider the inclusion of audio material, but also other communicative modes. The nonverbal part of the construct is given or should be given its own specific value in language learning and testing, as it becomes an essential part of the construct.

In the past few decades, research on FL listening skills has considered the type of input that is employed both in classroom teaching and for assessment purposes. Notably, the work of Gruba (1993, 1997, 2004, 2006), Vandergrift (1999, 2003, 2007, 2010, 2012), Vandergrift and Goh (2009, 2012) and Wagner (2007, 2008, 2010a, 2010b) examined how learners react to audio vs. video and/or multimedia sources, and the benefits and limitations of including one or more modes in listening tasks, designed to learn or test this skill. While some studies investigating the use of different input modes concluded that no difference is observed in the use of one mode (audio) as opposed to another (video listening) (Gruba, 1993; Coniam, 2001), other studies observed that learners show better results when watching video or multimedia material as opposed to listening to audio only versions of a listening test (Brett, 1997; Sueyoshi & Hardison, 2005; Wagner, 2010b). Learner performance in these tests is directly related to how we understand and interpret the listening text: only its verbal information, the nonverbal part of the text or both. Analyzing performance in multimodal tests without considering the types of nonverbal information implied in the listening process is basing experimental research on a theory that does not correspond to the listening

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construct. Gruba (1997) rightly pointed out that language comprehension tests must incorporate nonverbal communication aspects (e.g. culture or kinesics), which make test developers consider modes of presentations that include verbal, visual elements and also situational information, thus giving listening comprehension support to those test takers who are not particularly skillful in one single domain. This author advocates for the importance of video input in academic listening assessment and urges further research on this topic. A similar conclusion was found in Feak and Salehzadeh's (2001) study where they develop and validate an EAP video listening placement test. It could be said that all speakers make use or can make use of "visual symbol systems" to gather and interpret situational information in their L1, L2 or Ln, though it is true that the more information modes we may access, the more comprehension opportunities to understand a message we may have.

As Gruba (1997, p. 338) went on to state, "problems that may arise related to the influence of video on language proficiency assessment can be tackled within a larger perspective which sees visual elements as a key, not incidental, part of comprehending the world". It is this view of the "visual elements as a key" that needs to be taken into account for a more adequate approach to the multimodal listening construct. Thus, although these studies discussed in this section have contributed to the understanding of language learner listening skills and the use of different input media, in most cases some or all of the aspects below were not considered and need to be targeted in future studies, namely:

1. No pretests were carried out to determine students' overall proficiency level
2. No listening pretest was carried out to determine students' listening proficiency level
3. In most cases, no specific audio/video material and/or the questions used for the listening test employed in the study are included as part of the information facilitated in the article
4. Little importance is given to nonverbal communication in the listening input materials (i.e., whether there are significant intonation shifts, gestures, face expressions, etc. in the text that can be related to the test questions). An exception is Wagner (2010b) who briefly addressed this issue, and Vandergrift's (2011) analysis in interactive listening.
5. No information is given about whether any nonverbal skill was explicitly tested

10.3 Multimodal listening situations in formal learning settings

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This section presents a classification of the main situations or contexts in which listening can occur in formal learning settings at the university. Thus far, we have broadly distinguished two types of listening situations, transactional and interactional, attending to the level of interaction with the speaker. However, the complex construct of listening makes us consider other features that may shed some light on the demands of each academic listening situation to construct meaning, as well as their benefits in the multimodal listening learning process: 1) the type of setting, face-to-face or virtual; and 2) if communication takes place just mediated by some kind of technology (e.g. Massive Online Open Courses (MOOCs), Open Course Ware (OCW) platforms, Multimedia Educational Pills (MEPs), or/and by a lecturer in real time (e.g. lectures, webinars). Indeed, it is the situation that will provide us with a variety of modes that will, in turn, determine how the micro and macrostructure of the listening situation is organized. We will have to deal with different types of listening tasks and, accordingly, different assessment methods and assessment purposes, depending on:

1. whether we find ourselves in an interactional or transactional situation
2. the number of speakers participating and the kind of role they play (from more active to less active or passive)
3. the relationship among speakers (relative power among participants, social distance, etc.)
4. the kind of information we may need to elicit from this situation,.

In Figure 1 we summarize how we envisage the most significant listening situations that students may find in higher education, drawing on the level of interaction the students may have with the speaker in formal learning settings. As argued by Delanty (2001), the university has to play a new and crucial role in the 21st century knowledge society, though the adaptation to this changing scenario seems to be moving at slow pace. One step forward to fulfill part of the knowledge society's needs is represented by the growing offer of online undergraduate and postgraduate courses in online and traditional universities. Moreover, the presence of ICTs in face-to-face learning, as well as in blending learning courses, also demands special consideration since a multimodal nature is inherent to these resources.

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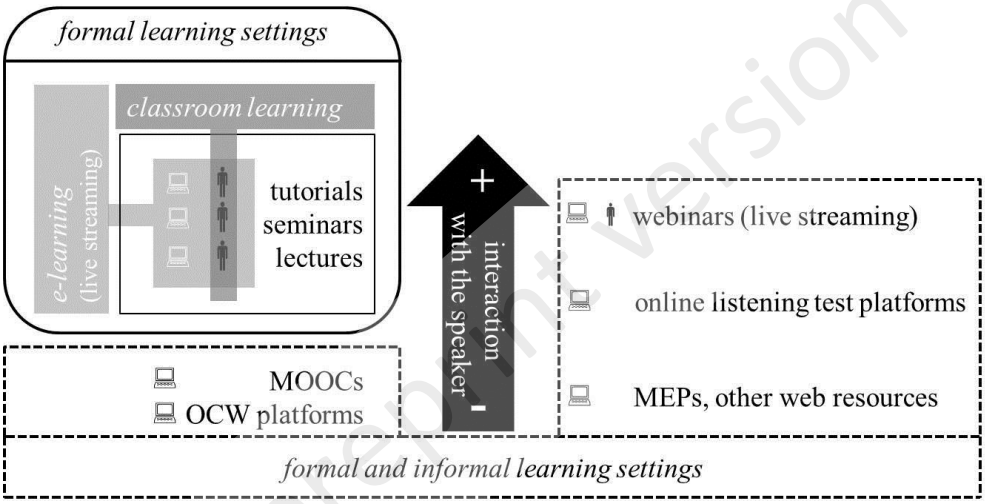

Figure 1. Multimodal listening situations in formal learning settings

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Let us consider each of the elements reflected in Figure 1. Formal learning may occur in a live setting (traditional, face-to-face setting) or in a virtual one. The e-learning setting varies in terms of how concerned the institution is in maintaining the lecturer's presence in a distance learning context. Regarding our main interest in this study, this is of special relevance for multimodal listening skills development. Whereas some universities with online courses choose to restrict interaction with the students to written communicative modes (emails, forums, or chats), others attempt to be closer to classroom learning and conferencing software is used to give lectures, seminars or tutorials (what is called live streaming). This is what we understand by an e-learning setting; there are clear differences from the classroom. However, the students also have the opportunity to listen to and watch the lecturers and PowerPoint presentations, or any other visual aid, in real time. In virtual delivery format, the visual input is commonly more reduced and focused on the speaker. Tutorials, seminars and lectures are the most frequent listening encounters in formal learning settings, and the degree of interaction is expected to decrease from tutorials to seminars and from seminars to lectures.

Regarding two-way interaction situations, we cannot ignore webinars and their important presence, both in formal and informal educational settings. This live stream resource serves, among others, the purpose of increasing international academic cooperation among universities. This gives students the opportunity to complete their learning by attending lectures from all around the world with minimum effort in terms of expenses and time. Students attending traditional courses find in webinars a different listening situation with

different interaction patterns. Though online students are familiar with this virtual listening context, usually they do not know the speaker. As a consequence, the aural and visual context and other features that will be discussed later are of paramount importance to decode the message and to construct the meaning in this situation.

Both listening and speaking skills can be used in two-way listening situations and assessment can also be done online. Using media like Skype teleconferencing would yield a more natural interpersonal context. Speaking skills are usually assessed in these situations, but there is less emphasis on listening ability with respect to what the student is able to comprehend when another speaker(s) is talking to him/her. Both or all speakers may use nonverbal communication devices. In this case, the speaker should not only be able to communicate nonverbally (encode nonverbal communication), but also to decode nonverbal communication conveyed by other speakers.

One-way listening situations are events where one person is giving some kind of public speech, for instance live academic lectures, or videoconferences that are increasing in popularity in graduate courses, and where participation of the listener is usually much lower than that of the main speaker of the event. Here multimodality applies most of all when the main speaker is a good communicator and makes ample usage of nonverbal communication. Likewise, MOOCs and OCW platforms give free access to video lectures, which are commonly used in informal learning, and can also be used in class as a videotext resource. Similarly, MEPs are mainly materials for transactional listening situations and can embed a considerable amount of nonverbal information.

In the different situations explained above, one or more modes may be salient. Listening sources may be: the teacher's or a speaker's voice, an audio text, a video text or multimedia texts. Audio and video input modes are defined by their main nature, that is, by whether the information conveyed is mostly auditory or visual. Thus, even though videotexts used in language teaching all include audio recording, it is the visual input that first comes to the mind when we think of using videotexts. If the added communication modes in videos as opposed to audios are the visual modes, when considering the use of videotexts in the language classroom the nonverbal visual information should be analyzed. This means that teachers need training in the interpretation of nonverbal cues that can be taught to students.

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When we talk about multimedia software, we intend the combination of different modes or media to express a message (Kress & van Leeuwen, 2001). The interest in finding out whether audio, video or multimodal resources are more suitable pedagogical devices for teaching listening skills lies in the assumption that user interaction with more than one communication mode may result in a more meaningful learning. Since an important feature of multimedia resources is their interactive nature, their use in language learning is essential when we want to pay attention to learner participation in the process. In this sense, it is not so much the teacher who decides the path a learner should follow, but it is the learner who may choose from a variety of paths offered by the teacher or by a specific multimedia product. One drawback for multimedia use in language learning is that all learners need to have or and be able to use a computer, while for watching a video or listening or a recording, only one electronic device is required and may be shared by all students. However, in this case intelligibility may be lower when a number of students listen or watch from screen or loudspeakers as opposed to using headphones or personal computers).

In most of these situations, listening implies watching. For most cases, events in which listening occurs are interactive and are therefore combined with speaking skills. It is for these two reasons that the teaching and assessment of listening skills should include the evaluation of other skills, i.e., multi-skill tasks should be fostered. Combining listening skills with other language skills may yield tasks that are more natural or similar to real life events and may thus be more meaningful for students who, in turn, could obtain better results. Imagine, for example, that we design a test that starts with a listening task in which learners pay attention to a video, including meaningful nonverbal information, and are later asked to perform an oral task that is similar to the video. Will students recall intonation patterns accompanied by gestures, and be able to reproduce them better when speaking in a parallel speaking task (i.e., parallel to the event they listened to)? The answer to this question demands further research on the benefits of using mixed skills tasks where one of the skills is multimodal listening. To a certain extent, this could be considered nonverbal information transfer/interpretation and application. That is, listening with a purpose beyond (linguistic) comprehension, that would lead to the design of meaningful learning situations for our students.

10.4. Multimodal listening instruction and assessment

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10.4.1 Monitoring listening and giving listening instructions

Systematic and purposeful observation of the listening practice is part of the monitoring process. Monitoring listening also considers the way activities progress towards specific listening aims. Thus, the way instructions are given is also important in the sense that tasks sequencing should be adequate both for the classroom tasks and for assessment situations. An important issue in these situations is the person who controls the listening task. The following is a list of the possible instructional and monitoring situations:

1. Teachers decide when, for how long and how many times an oral text is reproduced
2. Students interact with the audio or video recording by controlling it to achieve comprehension outcomes in a listening exercise. Students decide on the amount of time needed to achieve a goal within the given range of time for a task
3. Students interact with multimedia materials for listening comprehension practice in an autonomous manner with no time constraints and no testing on the part of the teacher

In a multimedia environment, the last two are employed, while in the classroom or in some testing situations, it is the first option that is frequently used. Nowadays, most students feel comfortable with a multimedia environment since they are quite familiar with it; it may even result in a calming effect as the students already have the images to rely on, so they can guess meanings even when they do not understand the words they hear.

Monitoring is an important piece of the listening task when trying to define how listening is taught and assessed within the listening construct. Monitoring takes place most of the time during classroom hours, especially for productive skills tasks. In multimodal listening assessment, being aware of the multiple sources of information is a complex task that needs the help of the teacher to clarify how to utilize the different modes and how these may combine. When teaching listening, careful monitoring of the students' progress in understanding multimodal patterns is necessary. Once these patterns are understood, students can monitor their multimodal pattern understanding. In fact, learners should be involved in comprehending the listening situation, interpreting the information, and trying to construct meaning from the various modes that are active in the multimodal listening. When students are assessed, the amount of control that they may have over the listening task may vary

depending on how we understand the task. Thus, in a multimodal environment, the degree of control and the interactional patterns need to be defined.

As indicated above, a difference is observed in the testing procedure regarding whether questions are answered while listening with no pauses, while listening with pauses or after listening. This is important particularly in multimodal environments. For example, if students are taking a test in multimedia format, the second option, i.e., listening and watching part one in an event and having some time to answer before the task continues to the second part, seems to be a more adequate administration procedure. The listener/watcher needs to process information in several modes, so keeping all that information in mind for all the questions until the end of the task is just not feasible especially in an exam with all the stress that maybe added in that situation.

Some of the research on listening tests, that compare audio and video input, criticize the use of video resources because students sometimes feel distracted by the video or they do not use it as frequently as expected to accomplish the listening activity (Coniam, 2001; Ockey, 2007; Wagner, 2007, 2010b). This is not surprising, but in this case, it is the design of the listening task that should be questioned: if the video does not provide useful information and the students are asked to answer questions while the video is on, then of course they will focus on the verbal information alone, or will find the video distracting. This aspect is also related to the issue of listening text types, which may refer to a specific presentation format and to a particular choice of mode (audio, video, multimedia), as well as the genre selected (lecture, interview, radio recording, etc.). Wagner (2010b), for instance, compared the effect of nonverbal information on test performance across different genres and concluded that nonverbal information affects performance. However, this study did not focus on when this nonverbal information is valuable per se, or in which cases the information enhances or clarifies the verbal information. Research is needed not only on whether visual information has a positive impact on performance, but also on how and when learners benefit from the inclusion of nonverbal information while listening.

10.4.2 Listening purpose and task format

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An important issue before deciding which listening format is the best to have a very clear idea of what we want to assess and how. Thus, task design and purpose should be thought our very carefully very closely. There are three main questions here that are relevant to reach a balanced proposal for multimodal learning objectives and assessment criteria:

1. The relationship between question type and modes employed in the listening test
2. The complexity of the listening task in terms of verbal and nonverbal information processing
3. The way test administration procedures are aligned with the question type and to the test's complexity (e-testing vs. paper)

The types of questions we decide to use in a listening test depend on the purpose of the listening task. The listening purpose may be contemplated from two perspectives: one is the kind of information that we want our students to retrieve from the listening test (context, verbal, gestural, etc.), and the other is what we want to ask them to do with that information.

Generally speaking, we listen to other people for a variety of purposes: to obtain information, to enjoy a conversation or someone's speech, to learn something in educational, professional or everyday contexts, or to try to simply understand someone or something. In the context of learning languages, the educational purposes may be summarized as follows (Nation & Newton, 2009; Weir, Vidaković & Galaczi, 2013):

1. Listening for accuracy: word or phrase comprehension
2. Listening to understand the main message or important information bits within a recording
3. Listening to understand complex information
4. Listening to understand detailed information or to find specific data. Information transfer (fill in chart, table while listening)
5. Listening to understand the context and to infer information

And in our multimodal proposal we would add:

6. Listening and *watching* to understand information from other modes different from verbal input

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The purpose of listening should determine the types of questions or tasks that are used and the administration procedures that are the most appropriate to answer those questions (i.e., while listening or after listening). There are different question formats that may be used for listening tasks (open or long answer; short answer; true/false; multiple choice; etc.). We need to consider how adequate each of these question formats can be for a listening task or test. Each format has its own advantages and disadvantages. For instance, multiple choice questions can be quite misleading since students are not asked if something they heard was true or false, but which option within several possibilities is true. This may be quite confusing for students since they are given more (misleading/complex) information. However, because multiple choice tests are easier to mark, they are a favored option. Designing true/false questions for a listening is an arduous task. In fact, many times deciding whether something is totally or partially true does not depend so much on understanding the message (more precisely, the words they hear) but on how we as individuals interpret that message. It is for this reason that answers that give a space for students to write a more or less long answer where they can really express their point of view are sometimes preferred. Written answers, however, are time consuming when marking exams and not cost effective, which is a drawback. At the same time, it is also true that multiple choice and true/false questions can be answered quickly in an exam situation, which brings us to our next reflection.

As we have seen in the previous section, we may use different text formats and different question types to assess listening skills. Yet if we include the ability to process more than one mode as part of the listening construct, the issue of purpose and question type becomes even more complicated. Each format may include one or more communicative modes. When just one mode is present, or when we want to focus on only one mode, we are working within a mono-modal approach. When two or more modes are active, we can talk about multimodality. Thus, depending on what we want to assess, we need to ask ourselves the following basic questions:

1. How do we assess listening comprehension, using individual or combined modes? That is, are we using a video to formulate…
    a. questions to assess the comprehension of audio/verbal information only,
    b. questions to assess visual information only,

c. a combination of both question types, or

d. questions using the combination of audio and video when they join to generate a cohesive message?

2. If several features belonging both to verbal and nonverbal communication are used at a specific point in a communicative event, what kind of assessment questions can we ask?

3. How can we formulate those questions so as not to confuse students?

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Listening tasks may also be formulated so that more than one skill is used. Thus, the purpose of the listening activity may be to use a mixed mode assessment question leading to the integration of two skills (Nation & Newton, 2009). For example, students may be asked to watch and listen to a video about a topic and then be requested to write an essay on that topic. In this case, we can talk about a mixed assessment mode, i.e., listening to retrieve information that will lead to a written task. Students may also be asked to listen to an audio recording of part of a telephone conversation in which a person is giving information on the different ways to reach a place. Then the student can be asked to inform a friend about how to reach that place. The second task focuses on the verbal message, while the first one allows for processing, interpreting, and evaluating verbal and nonverbal information to produce an outcome. In the task using multimodal input, the student can use verbal input, but also image input to create their written text. Imagine, for example, a video talking about a place which students may use their own interpretation of images or the verbal information provided in the video to describe that place. Similar tasks may be designed in combination with speaking skills.

For either types of tasks, very clear assessment criteria need to be defined both for the students and for the teachers and test raters in order to be able to discern whether the student has understood the input or part of it. Defining listening sub-skills is an important issue, particularly in this type of task where more than one main skill is required.

10.5 A framework for multimodal listening assessment

Our discussion on listening assessment, and particularly on using videotexts as the listening source, will focus on how students construct meaning using this multimodal input. We have

developed our proposal for multimodal listening assessment on the basis of Rost's (2011) model for listening assessment. He described the aspects of listening that are part of solid assessments as five types of knowledge: general, pragmatic, syntactic, lexical and phonological. The model we propose is seen from an even broader perspective to bring to the fore the largely ignored nonlinguistic components in listening assessment.

Nevertheless, to design our model it was important to consider the student's perspective in order to understand what components contribute to constructing meaning, and how this meaning is constructed. In Figure 2 we provide an overview of "the what and the how" of meaning construction when videotexts are used.

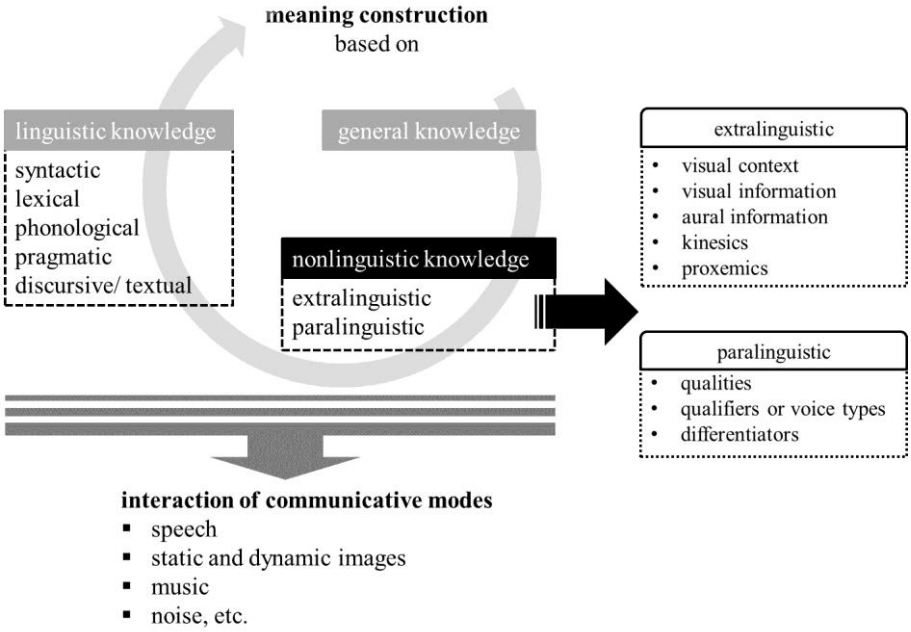

Figure 2. Meaning construction using videotexts

As can be seen, three types of knowledge should be part of the process of understanding videotexts: 1) general knowledge, 2) linguistic knowledge (which relates to any information that is decoded from linguistic input), and 3) nonlinguistic knowledge. The major change between our model and Rost's (2011) is the status given to extralanguage and paralanguage as components of nonlinguistic knowledge, which he considered under the umbrella of general knowledge. Our model also includes a fine grained description of extralinguistic knowledge, and also considers discursive and textual knowledge (inside linguistic knowledge) that serves to recognize genre and discourse patterns or rhetorical conventions.

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General and linguistic knowledge are commonplace in the description of listening skills and general language skills. In contrast, nonlinguistic knowledge needs further discussion in these contexts. As mentioned above, we understand this type of knowledge as articulated in extralinguistic and paralinguistic information. By extralinguistic knowledge, we refer to all the input the students receive from the videotext that has a visual or aural nature. Here we find five kinds of input: 1) visual context or visual background, 2) visual information (figures, tables, etc.), 3) aural information (music, background noise, etc.), 4) kinesics (gestures of hands and arms, head movement, facial expression, gaze, and body posture), and 5) proxemics, that is, interpersonal distance (Hall, 1959). With regard to paralinguistic knowledge, we follow Poyatos's (2002) definition of paralanguage that distinguishes three categories: 1) qualities (e.g. loudness, intonation rage, syllabic duration, etc.), 2) qualifiers or voice types, and 3) differentiators (e.g. laughter, throat-clearing, yawning, etc.).

Linguistic and nonlinguistic knowledge are ways of communication. Bara, Cutica and Tirassa (2001, p.73) claimed there are "superficial manifestations of a single communicative competence whose nature is neither linguistic nor extralinguistic, but mental". Even so, these are expressed physically with different modes to enable communication. The modes students find in videotexts can be speech, static and dynamic images, music, and noise, among others. Speech is primarily related to linguistic knowledge, but also to nonlinguistic one. In this respect, aural information (extralinguistic input), such as a background conversation, is speech, as is paralanguage. Static and dynamic images are part of the nonlinguistic knowledge. Static images are visual information, whereas dynamic images refer to kinesics and proxemics. Finally, music appears in many videotexts and its contribution to meaning has also been studied (Jewitt, 2003). Music, like noise, is another mode to express aural information.

If we acknowledge that all these features interact in multimodal listening, we should clearly describe them for students, including their study in the syllabus, as well as in test specifications within the listening construct. When working with a multimodal listening, we should ask students to interpret not only verbal information, but also to be able to identify, analyze and interpret all nonverbal modes of information. The study of gestures is one of the kinesic sub-modes that has received considerable attention from scholars (c.f. Bavelas,

Chovil, Lawrie, & Wade, 1992; Kendon, 2004; McNeill, 1992), together with facial expressions (c.f. Ekman, 2007). Gullberg (2006) discussed cross-cultural and cross-linguistic repertoires of gestures that should be considered as part of the acquisition of a language, in what she called "SLA of gestures", similarly to Raffler-Engel's (1980) "second kinesic acquisition".

In short, the construction on meaning in videotexts relies on understanding the complexity of the interaction of the modes that define them. Thus, the implementation of a multimodal listening assessment, understood also as part of the students' training for the development of their listening comprehension skills, could also consider some of the expected difficulties students may find due to the nature of videotexts. Among these, information density might be especially problematic, not only in terms of lexical density (as would occur in audio only listening), but also modal density (the linguistic and nonlinguistic interweaving of modes). Accordingly, in a situation in which there is a strong coherence and cohesion within and among modes, students would find less comprehension difficulties. But this is something that does not always occur in real life events. For example, Figure 3 depicts a complaint speech act in a video sequence1. The woman had asked her nephews to buy some bread while she was finishing lunch at home. But they found no bread at the bakery and simply went to the beach with friends. At lunch time there was no bread and she complains because they didn't even let her know there was no bread earlier in the morning. She says: "Why didn't you tell me?"

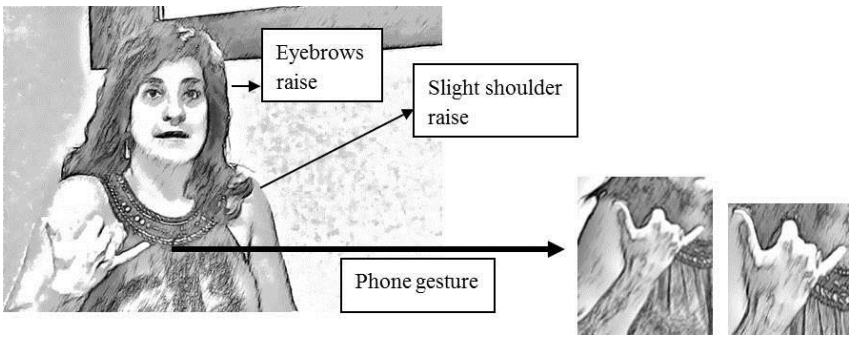

Figure 3. "Why didn't you tell me?"

The meaning she conveys goes beyond her words: they should have phoned her so she could have gone to the bakery later. The utterance employed does not mention any telephone use, an information that is exclusively understood if attention is paid to visual cues. The

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metaphoric gesture (a pictorial gesture that encodes an abstract idea, "phoning") shows she is reproaching them. Additionally, her feelings are also revealed by facial expression (raising eyebrows) and shoulders (raising shoulders slightly). This example illustrates important information conveyed by nonverbal input, which should also be considered when assessing listening comprehension, pushing students to focus on the whole input they receive in the videotext, rather than just on oral information. One question that could be asked in a listening activity could be "When they found no bread was left in the bakery, what should they have done?" a) phone their aunt, b) go back home, c) go to the beach".

One of the problems of testing listening is that, in many cases, there is no teaching practice as such, but rather the time devoted to listening skills in the classroom is basically also testing the skill, with little or no reflection on what the students listen to, why they listen, or how can students manage and improve their listening skills. Multimodal listening comprehension instruction should include comprehensible multimodal input. Likewise, practice in classroom listening should be parallel to the testing of this skill. It is for this reason that students should be given information as to the kinds of modes they should pay attention to, and classroom discussion should focus on all the available modes that come into action in the particular listening event to be practiced. At the same time, some instruction should appear in listening tests as to the modes the students should pay attention to before the listening starts.

It becomes obvious that both teacher and student training are needed in order to develop a multimodal working pedagogy. Figure 4 is a possible factsheet guide that students and teachers could work with in order to understand and evaluate multimodal texts in listening practice. It entails recognizing the multimodal video sequences or meaning units that comprise a specific event (the videotext listening comprehension). Here, listening sequences are understood as communicative units in multimodal discourse and for this reason become comprehension units when teaching multimodal listening skills.

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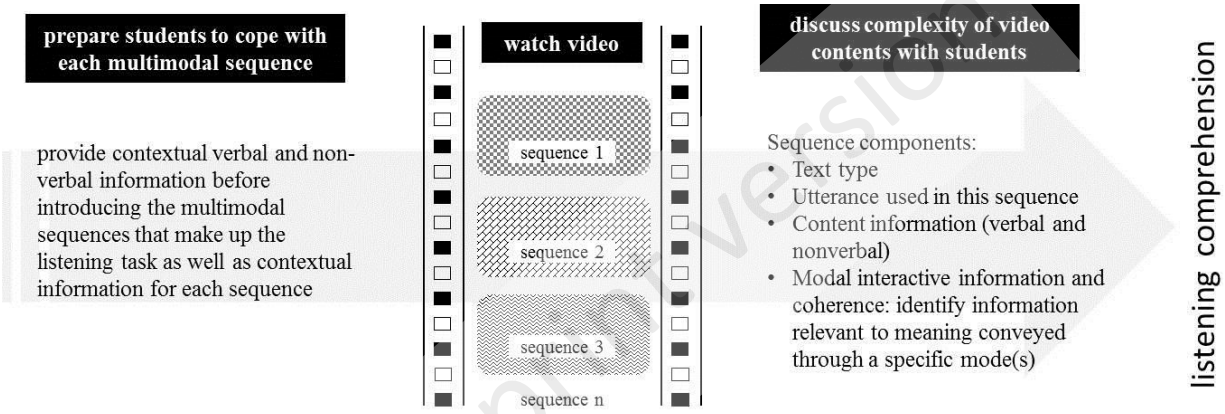

Figure 4. Information organization factsheet to address listening tasks

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In a testing situation, an important part of test design should be the presentation of a guiding set of questions, or brief information to situate the listener in the event in terms of general socio-cultural information. This should be done before the listening takes place in order to reduce cognitive burden. Expecting students to reach so many and varied knowledge types, as well as to follow the content of a listening activity in an exam setting, is perhaps putting too heavy a burden on them. In most formal listening tests, virtually no time is given for students to react to complex issues unless these are well-rooted in the students' previous knowledge, both cultural and pragmatic. Presenting context visuals, i.e., visual elements that provide information about the context (Ginther, 2002), as part of the pre-test questions could be considered not only a means to help students situate the event from a sociocultural point of view, but also to prepare them for the listening test.

10.6 Concluding remarks

Multimodal listening comprehension assessment should be able to measure the student's ability to identify, interpret and evaluate verbal and nonverbal information in multimodal texts in order to answer a set of questions or resolve a task. Nevertheless, interpreting nonverbal communication is not always an easy task as it typically depends on the context. Interpreting gestures on their own is not feasible; most gestures are polysemous and context-dependent.

Multimodal listening comprehension assessment should define criteria for evaluating the nonverbal aspects of multimodal listening. However, before that, relevant nonverbal

communication modes should be analyzed, synthesized and organized in a user-friendly way for students to be able to understand and deal with them in a coherent and productive way. The study of nonverbal communication and its specific importance in language learning must be prior to multimodal language assessment. Multimodal listening comprehension assessment also implies that teachers need to have clear principles on which to support their teaching practices.

Throughout this chapter, we have seen how analyzing and understanding the multimodal nature of listening to videos is a complex task that demands a great deal of effort from both students and teachers. The listening task per se becomes intricate. For this reason, further research into defining a difficulty threshold in multimodal listening would be useful. This could provide the grounds to enable the teacher or test rater to ascribe difficulty levels to language proficiency levels, helping teachers and test developers to select videotexts and design suitable tasks and activities accordingly.

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REFERENCES

Anderson, A., & Lynch, T. (1988). *Listening.* Oxford: Oxford University Press.

Ariza, E. N., & Hancock, S. (2003). Second language acquisition theories as a framework for creating distance learning courses. *The international review of research in open distance learning.* 4(2). Retrieved from: http://www.irrodl.org/index.php/irrodl/article/viewArticle/142/222

Bavelas, J.B., Chovil, N., Lawrie, D., & Wade, A. (1992). Interactive gestures. *Discourse Processes, 15*, 469-489.

Badger, R, Sutherland, P, White, G & Haggis, T (2001). Note perfect: an investigation of how students view taking notes in lectures. *System, 29*(3), 405-417.

Bara, B., Cutica, I., & Tirassa, M. (2001). Neuropragmatics: Extralinguistic communication after closed head Injury. *Brain and Language, 77*, 72-94.

Brett, P. (1997). A comparative study of the effects of the use of multimedia on listening comprehension. *System, 25*(1), 39-53.

Buck, G. (2001). *Assessing listening*. New York: Cambridge University Press.

Coniam, D. (2001). The use of audio or video comprehension as an assessment instrument in the certification of English language teachers: a case study. *System, 29*, 1-14.

Cross, J. (2011). Comprehending news videotexts: the influence of the visual content. *Language Learning & Technology, 11*(2), 44-68.

Delanty, G. (2001). *Challenging knowledge. The university in the knowledge society*. Buckingham / Philadelphia: Society for Research into Higher Education & Open University Press.

Ekman, P. (2007). *Emotions revealed. Recognizing faces and feelings to improve communication and emotional life*. (2nd edition). New York: Owl Books.

Feak, C., & Salehzadeh, J. (2001). Challenges and issues in developing an EAP video listening placement assessment. *English for Specific Purposes, 20*, 477-493.

Ginther, A. (2002). Context and content visuals and performance on listening comprehension stimuli. *Language Testing, 19*(2), 133-167.

Gruba, P. (1993). A comparison study of audio and video in language testing. *JALT Journal, 15*(1), 85-88.

Gruba, P. (1997). The role of video media in listening assessment. *System, 25*(3), 335-345.

Gruba, P. (2004). Understanding digitalized second language videotext. *Computer assisted language learning, 17*(1), 51-82.

Gruba, P. (2006). Playing the videotext: A media literacy perspective on video-mediated L2 listening. *Language Learning and Technology, 10*(2), 77-92.

Gullberg, M. (2006). Some reasons for studying gesture and second language acquisition (Hommage à Adam Kendon). *IRAL, 44*, 103-124

Hall, E. (1959). *The silence language*. Garden City, NY: Doubleday and Co.

Jewitt, C. (2003). Re-thinking assessment: multimodality, literacy and computer-mediated learning. *Assessment in education, 10*(1), 83-102.

Jewitt, C. (2013). Multimodal teaching and learning. In C.A. Chapelle (Ed.), *The encyclopedia of applied linguistics* (pp.4109-4114). Oxford: Blackwell Publishing Ltd.

Jung, E. (2003). The role of discourse signaling cues in second language listening comprehension. *Modern Language Journal, 87*(4), 562-577.

Kendon, A. (2004). *Gesture. Visible action as utterance*. Cambridge: Cambridge University Press.

Kress, G., & van Leeuwen, T. (2001). *Multimodal discourse. The modes and media of contemporary communication*. London: Edward Arnold.

Lynch, T. (2011). Academic listening in the 21st century: Reviewing a decade of research. *Journal of English for Academic Purposes, 10*, 79-88.

McGurk, H., & MacDonald, J. (1976). Hearing lips and seeing voices. *Nature, 976*(264), 746-748.

McNeill, D. (1992). *Hand and mind: what gestures reveal about thought*. Chicago & London: The University of Chicago Press.

Morell, T. (2004). Interactive lecture discourse for university EFL students, *English for specific purposes, 23*(3), 325-338.

Morell, T. (2007). What enhances EFL students' participation in lecture discourse? Student, lecturer and discourse perspectives. *Journal of English for Academic Purposes, 6*(3), 222-237.

Nation, I.S.P. & Newton. J. (2009). *Teaching ESL/EFL listening and speaking*. Routledge: New York.

Ockey, G. (2007). Construct implications of including still image or video in computer-based listening tests. *Language Testing 24*(4), 517-537.

preprint version

Poyatos, F. (2002). *Nonverbal communication across disciplines. Volume II. Paralanguage, kinesics, silence, personal and environmental interaction*. Amsterdam: John Benjamins.

Raffler-Engel, W. V. (1980). Kinesics and second language acquisition. In B. Kettemann & R. N. St. Clair (Eds.), *New approaches to language acquisition* (pp. 101–109). Tübingen: Gunter Narr Verlag.

Rost, M. (2011). *Teaching and researching listening.* Edinburgh: Pearson Education Limited.

Shellenbarger, S. (2014, July 22). Tuning in: improving your listening skills. How to get the most out of a conversation. *The wall street journal.* Retrieved from: http://online.wsj.com/articles/tuning-in-how-to-listen-better-1406070727?tesla=y&mg=reno64-wsj&url=http://online.wsj.com/article/SB10001424052702304058404580045441119703492.html

Sueyoshi, A., & Hardison, D. M. (2005).The role of gestures and facial cues in second language listening comprehension. *Language learning, 55*, 661-699.

Tauroza, S., & Allison, D. (1994). Expectation-driven understanding in information systems lecture comprehension. In J. Flowerdew (Ed.), *Academic listening: Research perspectives* (pp. 35-54). Cambridge: Cambridge University Press.

Vandergrift, L. (1999). Facilitating second language listening comprehension: acquiring successful strategies. *ELT Journal, 53*(3), 168-176.

Vandergrift, L. (2003). Orchestrating strategy use: Toward a model of the skilled second language listener. *Language learning, 5*(3), 463-496.

Vandergrift, L. (2007). Recent developments in second and foreign language listening and comprehension research. *Language teaching, 40*, 191-210.

Vandergrift, L. (2010). Researching listening in applied linguistics. In B. Paltridge & A. Phakiti (Eds.), *Companion to research methods in applied linguistics* (pp. 160-173). London: Continuum.

Vandergrift, L. (2011). L2 listening: Presage, process, product and pedagogy. In E. Hinkel (Ed.), *Handbook of research in second language teaching and learning,* Volume II (pp. 455-471). New York: Routledge.

Vandergrift, L. (2012). Teaching interactive listening. In H. P. Widodo & A. Cirocki (Eds.), *Innovation and creativity in ELT methodology* (pp. 1-14). New York: Nova Science Publishers.

preprint version

Vandergrift, L., & C. Goh, (2009). Teaching and testing listening comprehension, In Long, M., & Doughty, C. (Eds.) *Handbook of language teaching*. Malden, MA: Blackwell.

Vandergrift, L., & Goh, C. (2012). *Teaching and learning second language listening: Metacognition in action.* New York: Routledge.

Wagner, E. (2007). Are they watching? Test taker viewing behavior during an L2 video listening test. *System, 11*(1), 67-88.

Wagner, E. (2008). Video listening tests: what are they measuring? *Language assessment quarterly, 5*(3), 218-243.

Wagner, E. (2010a). Test-takers' interaction with an L2 video listening test. *System, 38*, 280-291.

Wagner, E. (2010b). The effect of the use of video texts on ESL listening test-taker performance. *Language testing, 27*(4), 493-513.

Weir, C., & Vidaković, I. (2013). The measurement of listening ability 1913-2012. In C. Weir, I. Vidakovićc & E. Galaczi (Eds.) *Measured constructs* (pp. 347-419). Cambridge: Cambridge University Press.

NOTES

This was a real conversation the authors witnessed and have reproduced for this chapter.

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